

AI-Based Fake Identity Document Screening System

Two-Stage Verification Pipeline with Multi-Signal Document-Type Validation, Client-Side OCR, Transfer-Learning Computer Vision & Tampering Detection

A comprehensive production-grade document verification system specifically engineered to detect fraudulent, manipulated, and mismatched identity documents (Aadhaar Card, PAN Card, Passport, and Voter ID). Features a decoupled multi-signal Stage 1 document classification engine, an isolated Stage 2 authenticity verification model, automated client-side forensics, and an integrated AI fraud explanation assistant.

Target Documents	Aadhaar Card, PAN Card, Indian Passport, Voter ID
System Architecture	Two-Stage Decoupled Multi-Signal Pipeline
Core Technologies	TensorFlow.js, Tesseract.js, MobileNetV2, Express.js, Gemini Vision LLM
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1. Executive Summary & Problem Statement

1.1 Executive Summary

Digital identity verification is a fundamental pillar of modern banking (KYC), immigration, government services, and online platforms. With the widespread accessibility of generative image tools, photo-editing software, and synthetic identity fraud, organizations face an unprecedented volume of forged and counterfeit identity documents.

This project delivers an intelligent, end-to-end, multi-stage identity verification platform capable of screening Indian identity cards—specifically **Aadhaar Card, PAN Card, Passport, and Voter ID Card**. Unlike conventional verification systems that blindly execute a single binary classification model, our architecture implements a **Two-Stage Validation Pipeline** that guarantees document-type identity before assessing authenticity.

1.2 The "Cross-Type False Confidence" Vulnerability

Earlier generation computer vision classifiers suffer from a critical algorithmic flaw known as *Cross-Type Confidence Inflation*. When separate binary classifiers are trained exclusively on one document type (e.g. an Aadhaar model trained solely on original vs. fake Aadhaar images):

The Problem: When a user selects "Aadhaar Card" and uploads a completely different document (such as a Passport or Voter ID), the Aadhaar binary model—never having seen a Passport during training—still forces a softmax probability distribution over [ORIGINAL, FAKE]. This results in absurd verdicts like "Aadhaar ORIGINAL 99.8%" on a clear Passport image.

Furthermore, selecting the highest score among all binary models to infer the document type is equally flawed: because each binary model outputs ~99-100% confidence for almost any input, model confidence cannot be used for type identification.

1.3 Solution Overview: Decoupled Two-Stage Architecture

To permanently resolve this vulnerability, our system strictly enforces a decoupled architecture:

- **Stage 1 (Document Type Validation):** Evaluates three independent signals (Multilingual OCR, Dedicated 5-Class Doctype Classifier, and AI Vision LLM) under a strict priority hierarchy to establish *what* the document is. Per-type authenticity models are **strictly prohibited** from participating in Stage 1.
- **Stage 2 (Authenticity & Quality Verification):** Only executes once Stage 1 confirms that the uploaded document matches the selected document type. If a mismatch or unrecognized document is detected, Stage 2 scores are completely suppressed and hidden.

2. End-to-End System Architecture

The system consists of a high-performance client-side frontend served via Python HTTP (Port 5000) and a secure Node.js Express backend proxy (Port 5001) that handles external multimodal API calls without exposing client keys.

2.1 High-Level Architectural Flowchart

```
+-----+
+-----+ | USER INPUT & ACQUISITION | | [ File Upload: JPG, PNG, WEBP ] OR [ Real-Time Webcam
Video Stream ] | +-----+
+-----+ | v +-----+
+-----+ | STAGE 1: DOCUMENT TYPE VALIDATION ENGINE (Decoupled) |
| | | [ SIGNAL 1A: Multilingual OCR ] [ SIGNAL 1B: Doctype Model ] [ SIGNAL 1C: AI Vision LLM
] | | - Tesseract.js (eng+hin+kan) - MobileNetV2 (5 classes) - Gemini Vision LLM | | - Scale
≥ 1600px + Gamma boost - Letterbox 224x224 input - Visual layout inspection | | -
0/90/180/270 deg rotation - 4-way rotation averaging - Security emblem analysis | | - Regex &
Levenshtein matching - Strict conf threshold ≥0.85 - Tampering detection | +-----+
+-----+ | Priority
Decision Arbitration (Rules 1-5) | +-----+
+-----+ | | [ Type Confirmed ] [ Mismatch / Unrecognized ] | | v v +-----+
+-----+ +-----+ | STAGE 2: AUTHENTICITY ENGINE | | STAGE 1
WARNING SHIELD | | - Load Selected Binary Model | | - Suppress Stage 2 Verdict | | - Letterbox
& Rotation Average | | - Hide ORIGINAL/FAKE scores | | - Forensic Quality Checks: | | -
Display Mismatch Banner | | * Laplacian Blur Variance | | - Offer 1-Click Type Switcher | | *
Exposure & Luminance Bounds | +-----+ | * Screen Moiré Pattern
Check | | - Verdict Synthesis: | | ORIGINAL | FAKE | SUSPICIOUS | +-----+
+-----+ | v +-----+
+-----+ | OUTPUT & INTERACTIVE REPORTING | | [ Visual Verdict Card ] [ Multi-
Signal Breakdown ] [ Interactive Forensic AI Chatbot ] | +-----+
+-----+
```

2.2 Component Roles & Separation of Concerns

Layer / Component	Host & Port	Key Responsibility
Web UI & Canvas Engine	Client (Port 5000)	Image ingestion, live webcam stream, aspect-ratio letterboxing, quality metrics computation, UI state rendering.
Client-Side OCR (Tesseract.js)	Web Worker (CDN)	Multi-language OCR processing in background threads without server network round-trips.
Dedicated Doctype Model	Local Client / WebGL	5-class MobileNetV2 inference executed in client browser memory via TensorFlow.js.
Binary Authenticity Models	Local Client / WebGL	Per-document-type Teachable Machine models trained exclusively on original vs fake samples.
Backend Proxy Server	Node.js (Port 5001)	Provides <code>/api/verify-doc-type</code> , <code>/api/vision-status</code> , and <code>/api/explain-fake</code> while securely safeguarding the Gemini API key.

3. Stage 1: Document Type Detection Engine

Stage 1 acts as an impenetrable gatekeeper. It evaluates the uploaded document against three independent signals and arbitrates through a deterministic 5-rule decision hierarchy.

3.1 Signal 1A: Enhanced Multilingual OCR Engine

Optical Character Recognition serves as the most definitive proof of document identity. Real-world documents are often photographed in suboptimal lighting, low resolutions, or rotated orientations. To overcome this, our system executes a 4-step image enhancement pipeline:

- 1. Long-Edge Upscaling:** Images with dimensions below 1600px are bilinearly upscaled to a target long edge of at least 1600px (capped at 2200px) to maximize OCR character glyph recognition.
- 2. Luminance & Gamma Brightening:** The average luminance $L_{\text{avg}} = \frac{1}{N} \sum (0.299R + 0.587G + 0.114B)$ is computed. If $L_{\text{avg}} < 80$, a non-linear gamma expansion ($\gamma = 0.55$) is applied to boost dark shadows without blowing out highlights: $L_{\text{out}} = 255 \times \left(\frac{L_{\text{in}}}{255}\right)^\gamma$
- 3. Adaptive Contrast Stretching:** A piecewise contrast enhancement expands the dynamic range around midtones.
- 4. Four-Angle Orientation Sweep:** The engine tests $0^\circ, 90^\circ, 180^\circ,$ and 270° . If a strong identifier is detected at any angle, the sweep terminates early to optimize latency.

OCR Keyword & Regex Match Hierarchy

Doc Type	Strong Keywords	Regex Pattern	Weak Keywords
Aadhaar Card	uidai , unique identification authority , aadhaar , आधार , आदर्श	<code>\b[2-9]\d{3}\s?\d{4}\s?\d{4}\b</code> (12 digits)	Government of India, DOB, Male, Female
PAN Card	income tax department , permanent account number	<code>\b[A-Z]{5}\d{4}[A-Z]\b</code> (PAN structure)	Govt. of India, Father's Name, Signature
Passport	passport no , Compound: nationality + expiry	<code>\bP<IND[A-Z0-9<]{10,}\b</code> (ICAO MRZ)	Republic of India, Given Name, Surname
Voter ID	election commission of india , भारत निर्वाचन आयोग	<code>\b[A-Z]{3}\d{7}\b</code> (EPIC Number)	Elector's Name, EPIC, Gender, Age

3.2 Signal 1B: Dedicated Document Type Classifier (./model-doctype/)

When OCR text is unreadable or blurred, the system relies on a dedicated 5-class MobileNetV2 neural network. Trained across five distinct classes (aadhaar , pan , passport , voter , other), this model evaluates macro-level visual features: layout geometry, color palettes, photo positioning, and structural emblems.

- **Letterbox Preprocessing:** Maintains image aspect ratio by embedding the document inside a square 224×224 canvas with neutral gray padding (#808080), preventing geometric distortion.
- **Rotation Invariance:** Evaluates the image across four 90° rotations and averages the softmax probability vectors:
$$P(c) = \frac{1}{4} \sum_{\theta \in \{0, 90, 180, 270\}} P_{\theta}(c)$$

3.3 Signal 1C: Multimodal AI Vision LLM (Gemini Vision)

For complex edge cases (e.g. heavily damaged or obscured documents), the system calls the backend endpoint `/api/verify-doc-type`. The multimodal model inspects layout symmetry, governmental emblems, photo borders, and security watermarks, returning a structured JSON response containing:

```
{
  "type": "aadhaar" | "pan" | "passport" | "voter" | "other",
  "confidence": 0.0 - 1.0,
  "reason": "Clear presence of UIDAI emblem and 12-digit number format.",
  "fake_signals": ["screen_moire_detected", "font_inconsistency"]
}
```

3.4 Priority Decision Rules (Rules 1 to 5)

1. **Rule 1 (Definitive OCR Match):** If OCR yields strong identifiers for exactly ONE document type, that type is immediately selected with 99% confidence. All subsequent signals are bypassed.
2. **Rule 2 (Multiple OCR Matches):** If multiple types return strong keywords, the type with the highest score is chosen, but flagged as *Low Certainty* (82% confidence), offering user confirmation buttons.
3. **Rule 3 (Dedicated Doctype Classifier):** If OCR yields no strong hits, the neural classifier is inspected. If its top probability is ≥ 0.85 and no competing rival class exceeds 0.30 , its verdict is accepted.
4. **Rule 4 (AI Vision LLM):** If the local classifier is ambiguous or missing, the Vision LLM is queried. If confidence ≥ 0.75 , its prediction is adopted.
5. **Rule 5 (Fallback / Unrecognized):** If all signals fail or point to `other`, the system triggers the *Unrecognized Document* shield.

4. Stage 2: Authenticity Screening & Forensic Analysis

Stage 2 is executed exclusively when Stage 1 has confirmed that the uploaded document matches the user's selected category. It combines deep neural feature extraction with client-side image forensics.

4.1 Strict Isolation Policy

Security Guarantee: Whenever Stage 1 detects a document-type mismatch or an unrecognized image, the entire Stage 2 results panel (including the ORIGINAL/FAKE progress bars and confidence percentages) is completely hidden. The system replaces it with the message: `Not verified - document type unconfirmed`.

4.2 Forensic Image Quality Checks

Even genuine documents may fail if the image is captured in poor conditions. The system runs real-time forensic checks directly on the HTML5 Canvas pixel buffer:

70.0

LAPLACIAN BLUR
THRESHOLD

35 - 225

LUMINANCE EXPOSURE
BOUNDS

400×300

MINIMUM RESOLUTION
(PX)

FFT Moiré

SCREEN CAPTURE
DETECTION

Algorithmic Details:

- Laplacian Variance Blur Detection:** Convolves a 3×3 discrete Laplacian operator over the grayscale image to measure high-frequency edge energy. A variance below 70.0 indicates camera motion blur or out-of-focus capture:
$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$
- Exposure & Luminance Analysis:** Average pixel brightness must fall within 35 to 225. Values below 35 represent severe underexposure; values above 225 represent overexposure and washed-out text.
- Screen Moiré / Recapture Detection:** Computes high-frequency diagonal periodic pixel intensity variances. Documents captured off an LCD/OLED monitor exhibit characteristic moiré interference fringes.

4.3 Verdict Synthesis Logic

Verdict	Trigger Condition	Action & Recommendation
ORIGINAL	Original probability $> 50\%$ and no quality or tampering flags.	Accepted. Document verified authentic.
FAKE	Fake probability $\geq 50\%$ OR ≥ 2 structural tampering flags.	Rejected. Potential forgery or digital manipulation detected. Activates AI Explanation Chatbot.
SUSPICIOUS	Model vs. AI visual discrepancy OR display moiré detected.	Flagged for manual compliance officer review.
LOW CONFIDENCE	Model predicts ORIGINAL but image is blurry, dark, or low-res.	Prompt user to recapture document in better lighting.

5. Datasets, Model Training & Class Balancing

5.1 Extracted Training Datasets

The model training datasets were aggregated and structured from AI-based fake identity document system250 :

Document Class	Original Folder Path	Originals	Fakes	Total Base Samples
Passport	Passport original / Passport fake	34	240	274 images
PAN Card	original pancard / fake pancard	106	11	117 images
Voter ID	original voterID / fake voter ID	12	174	186 images
Aadhaar Card	original adhar / fake adhar + samples	3	6	9 images
Other (Non-Doc)	Screenshots, Wallpapers, Synthetic Cards	—	—	200 images

5.2 Data Preprocessing & Deduplication

- Deduplication via Hash:** Every input file is hashed using SHA-256 over raw pixel bytes to eliminate redundant downloads and augmentations.
- 80 / 10 / 10 Split:** Data is split by file into 80% Training, 10% Validation, and 10% Test partitions. Stratification guarantees that minority classes (such as Aadhaar) have samples represented across all three splits.
- Aadhaar Minority Class Balancing:** Because Aadhaar has a smaller base sample count (9 images), heavy offline and online data augmentation is applied:
 - Random rotations: $\pm 25^\circ$
 - Brightness jitter: $0.7\times$ to $1.3\times$
 - Random zoom: $0.8\times$ to $1.2\times$
 - Color jitter & slight perspective shear

Furthermore, balanced class weights are injected into the loss function: $W_c = \frac{N_{\text{total}}}{C \times N_c}$ ensuring that the gradient update from each Aadhaar image is proportional to its representation.

5.3 Neural Architecture & Web Export

The model utilizes **MobileNetV2 ($\alpha = 0.35$)** with ImageNet transfer learning weights. The feature extractor produces a 1280-dimensional embedding vector, followed by:

```
Sequential [  
  MobileNetV2 Base (alpha=0.35, input=[224, 224, 3]) → 1280-dim feature vector  
  Dense Layer 1 (units=100, activation='relu', use_bias=true)  
  Dense Layer 2 (units=5, activation='softmax', use_bias=false)  
]
```

The trained model is serialized into standard Teachable Machine / TensorFlow.js format:

- `model.json` : Contains full model topology and weights manifest.
- `weights.bin` : Contiguous float32 little-endian binary weights buffer.
- `metadata.json` : Contains `{"modelName":"tm-my-image-model","labels":["aadhaar","pan","passport","voter","other"],"imageSize":224}` .

6. User Interface, Interactive Chatbot & Debug Mode

6.1 Glassmorphic User Interface

The web frontend is designed with a modern dark-mode aesthetic utilizing CSS variables, responsive CSS grid layouts, and hardware-accelerated canvas transforms. Key elements include:

- **Dynamic Document Selector:** Tab buttons for Aadhaar, PAN Card, Passport, and Voter ID with instant state updates.
- **Dual Ingestion Modes:** Drag-and-drop file upload zone alongside full webcam integration supporting live image capture.
- **Confidence Display:** All displayed confidences are capped at 99% to prevent misleading 100% assertions.
- **Doctype Missing Warning Banner:** Automatically alerts administrators on startup if `./model-doctype/` is not found:

Document-type model missing: Run `train_doctype.py` to build the dedicated 5-class model.

6.2 Real-Time Debug Inspector (‘?debug=1’)

Engineers and auditors can inspect internal pipeline signals by toggling the debug inspector (via the header button or appending `?debug=1` to the URL). The debug panel reveals:

Signal Field	Reported Information
Final Decision & Rule	Shows which specific rule (Rule 1 to 5) determined the document identity and with what confidence.
OCR Confidence & Angle	Displays Tesseract mean confidence score (0-100%) and the rotation angle (0° , 90° , 180° , 270°) that produced the match.
OCR Raw Text Extracted	Provides the exact first 300 characters of raw OCR text for immediate inspection of spelling and artifacts.
Per-Document OCR Table	Displays counts for regex pattern hits, strong keyword hits, weak hits, and total calculated score per document type.
Doctype Model Probabilities	Displays the raw softmax probability distribution across all 5 classes (<code>aadhaar</code> , <code>pan</code> , <code>passport</code> , <code>voter</code> , <code>other</code>).
Vision AI LLM Status	Reports whether Gemini Vision is <code>Online</code> , <code>Optional - Not Configured</code> , or <code>Offline</code> , along with reasons and tampering flags.

6.3 Interactive Forensic AI Chatbot

When a document is flagged as **FAKE** or **SUSPICIOUS**, the system automatically opens an interactive slide-in chat drawer. Powered by the Express proxy calling Google Gemini:

- **Structured Initial Briefing:** Outlines 3–5 bullet points detailing specific red flags typical of that document type (e.g. font misalignment, photo edge tampering, missing micro-printing, or corrupted QR codes).
- **Strict Persona Enforcement:** The assistant strictly adheres to document security topics, never claiming to examine invisible pixels and politely redirecting off-topic queries.
- **Contextual Follow-Up:** Users can ask follow-up questions regarding how to distinguish authentic watermarks, verify MRZ checksums, or obtain genuine duplicates.

7. Operational Guide & System Verification

7.1 Local Deployment & Port Mapping

The system is designed to run locally with zero cloud dependencies for Stage 1A, Stage 1B, and Stage 2:

```
# 1. Start Client-Side Web Application Server (Port 5000)
cd "c:\...\AI-Based-Fake-Identity-Document-Screening-System-main"
python -m http.server 5000

# 2. Start Secure Express Backend Server (Port 5001)
cd server
npm install
node server.js
```

7.2 Environment Configuration (`.env`)

To enable multimodal AI vision and interactive chatbot features, configure the backend `server/.env` file:

```
PORT=5001
GEMINI_API_KEY=your_gemini_api_key_here
```

If `GEMINI_API_KEY` is omitted, the system operates seamlessly in offline mode: Signal 1A (OCR) and Signal 1B (Local Doctype Model) perform Stage 1, while canned forensic responses support the chatbot.

7.3 System Verification & Test Matrix

Test Scenario	Uploaded Document	Selected Type	Expected System Behavior	Status
Correct Match	Real Aadhaar Card	Aadhaar Card	Stage 1 confirms Aadhaar via Rule 1 (OCR) → Stage 2 evaluates authenticity → Verdict: ORIGINAL.	PASS
Type Mismatch	Passport Image	Aadhaar Card	Stage 1 detects Passport → Stage 2 suppressed → Mismatch Banner: <i>"Selected Aadhaar, looks like Passport"</i> → 1-click switch button displayed.	PASS
Forged Document	Tampered PAN Card	PAN Card	Stage 1 confirms PAN Card → Stage 2 neural model flags manipulation → Verdict: FAKE → AI Chatbot drawer activates.	PASS
Non-Document	Selfie / Wallpaper	Voter ID	Stage 1 flags Unrecognized / Other via Rule 5 → Stage 2 suppressed → Warning: <i>"Unrecognized Document"</i> .	PASS
Dark / Blurry	Poor Lighting Passport	Passport	Stage 1 upscales & boosts gamma → Type recognized → Stage 2 Laplacian variance < 70.0 → Verdict: LOW CONFIDENCE.	PASS

7.4 Summary & Conclusion

This AI-Based Fake Identity Document Screening System successfully eliminates the cross-type misclassification vulnerability inherent in traditional single-stage screening architectures. By decoupling document identification (Stage 1) from authenticity classification (Stage 2), applying multi-signal priority arbitration, and enforcing real-time quality forensics, the system delivers an enterprise-grade, transparent, and resilient document verification platform.